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Simulation of the BEAVRS benchmark using VERA ☆

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ABSTRACT

The Virtual Environment for Reactor Applications (VERA) is being developed by the Consortium for the Advanced Simulation of Light Water Reactors (CASL). VERA is used to perform the Benchmark for Evaluation and Validation of Reactor Simulations (BEAVRS), which provides two cycles' worth of operating power history, along with a full, detailed description of the geometry, and measured data. Cycle 1 and 2 are simulated with VERA and the results are compared to the measured zero power physics tests, critical boron concentration, and flux maps.

1. Introduction

Development of a core simulator capability in VERA (Turner et al., 2016) is crucial to achieving the goals of CASL (CASL, 2019). VERA must be capable of accurately predicting the detailed power, temperature, and isotopic distribution in the reactor throughout the lifetime of the fuel. This information becomes the basis for understanding the underlying challenge problems that CASL is tasked to address. This work focuses on using VERA to analyze the Benchmark for Evaluation and Validation of Reactor Simulations (BEAVRS) (Horelik et al., 2013) for comparisons with measured plant data.

The BEAVRS Benchmark was released by MIT in 2013 to provide data from an operating nuclear reactor to the public to allow for validation of methods developments. This benchmark is similar to the data provided in the VERA Progression Benchmark (Godfrey, 2014). The BEAVRS benchmark provides two cycles' worth of operational history, including power levels and boron concentrations. In addition, flux maps are provided at several points during each operating cycle.

2. Modeling methodology

The state-of-the-art capabilities within VERA provide unprecedented resolution for reactor analysis through high-fidelity multi-physics couplings. The simulator components for steady-state reactor core simulation have been selected to eliminate the barriers facing modern industrial methods for improved accuracy on smaller spatial scales. VERA provides direct, fully coupled solutions at the fuel rod level for neutronics and thermal-hydraulics without any spatial homogenization. Isotopic depletion and transmutations occur locally within the once-through 3-D calculation, avoiding the need for macroscopic spectral corrections to simplified history models. The user interface is designed for ease of use and provides a single common geometry model to each of the underlying physics codes. VERA also manages the calculation flow, data transfer, and convergence between methods automatically, and it is capable of computational scaling from leadership class supercomputers to engineering-grade compute clusters, enabling access for scientists and engineers across many industrial application areas. The individual physics methods employed by this application of VERA are described in the following sections.

2.1. MPACT transport solver

The 3D pin-resolved reactor transport code MPACT (MPACT, 2015) has been under development by Oak Ridge National Laboratory (ORNL) and the University of Michigan. The methods in

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MPACT have been documented in several sources (Collins et al., 2016; Kochunas et al., 2016); a brief outline is presented here that discusses the 3D solution mechanism using the 2D/1D method and detector models. The 2D/1D method is used to solve the neutron flux distribution throughout the core. This is accomplished by using the 2D method of characteristics (MOC) in the radial planes to capture the heterogeneity in the radial direction with high accuracy. Each pin cell is explicitly modeled, and even sub-pin detail can be captured. In the axial direction, a low-order transport solution is obtained through NEM-P_3 on a pin-cell homogenized basis. The axial and radial solutions are linked through the use of transverse leakage terms which ensure neutron balance in every pin cell at convergence. More detail on the 2D/1D methodology used in MPACT can be found in Collins et al. (2016). MPACT uses a 51 energy group cross section library (Kim et al., 2017) based on ENDF/B VII.1 data with subgroup parameters to capture self-shielding effects. MPACT can estimate the detector response during cycle operation for comparison with raw flux map data. This is accomplished by explicitly modeling the instrument tubes in the geometry. Instead of placing a trace amount of U-235 into the instrument tube during the simulation, MPACT calculates the detector response during a post processing step. Once the flux solution is obtained, the local flux in each instrument tube is folded together with the U-235 cross section from the cross section library at the local temperature. This signal is obtained in every instrument tube and at every axial level in the model. The data are normalized and written to a binary output file for post-processing comparisons to the measured data.

In addition to solving the transport equation, MPACT performs many other key capabilities in the simulation. These capabilities include:

- Critical boron search
- Equilibrium boron search
- Predictor-corrector time stepping algorithm directly coupled to the Oak Ridge Isotopic Generation (ORIGEN) code (Gauld et al., 2011)
- Special treatment for control rods to prevent decusping effect (Graham et al., 2018)
- Ability to read and write an isotopic restart file
- Ability to shuffle fuel between cycles and decay the isotopes during the outage.

All of these capabilities are essential to performing practical core analysis using VERA.

2.2. COBRA-TF thermal hydraulic solver

COBRA-TF (CTF) is a modernized, improved, quality-controlled version of the COBRA-TF (Salko and Avramova, 2019) subchannel thermal-hydraulic code. The code is being developed and maintained by Oak Ridge National Laboratory and North Carolina State University as part of the CASL program. In VERA, CTF is directly coupled to MPACT and is executed in full for each neutronics-T/H iteration until convergence is reached between the two codes.

CTF uses a transient two-fluid three-field (i.e. liquid film, liquid drops, and vapor) modeling approach. A wide range of flow-regime-dependent closure models are available for capturing complex two-phase flow behavior, which includes rod-to-fluid heat transfer, interphase heat and mass transfer, wall and interphase drag, turbulent mixing and void drift, grid droplet breakup, and grid heat transfer enhancement effects. The rod-to-fluid heat transfer models were designed to handle the entire boiling curve, including single-phase flow, subcooled and bulk boiling, critical heat flux, and post-critical heat flux heat transfer. CTF has been used in many applications before and during the CASL program, including: modeling single-phase normal operating conditions, modeling two-phase flow in accident conditions, modeling of boiling water reactors (BWR), uncertainty quantification, and benchmarking activities. A higher-fidelity model of the core is achieved with CTF by individually modeling each rod-bundle coolant channel and pin in the core.

2.3. Isotopic depletion

The Oak Ridge Isotope Generation (ORIGEN) depletion/decay code was developed at ORNL in 1973. In 1980, the development diverged into two versions, ORIGEN2 and ORIGEN-S, with ORIGEN2 intended for stand-alone depletion/decay calculations, and ORIGEN-S intended for coupled neutron transport/depletion/decay analysis within SCALE (SCALE, 2011). Since the divergence, many new versions have been created, most of which are modified versions of ORIGEN2 and are readily available from the Radiation Safety Information Computational Center (RSICC). In 1989, support for ORIGEN2 was cancelled. Since 1989, ORIGEN-S, recently rebranded simply as “ORIGEN”, is the only variant under active development and since the inception of the original ORIGEN in 1973, it has been the only version supported by ORNL, with continuous updates to data and methodology as part of the SCALE code system.

ORIGEN has been used to model nuclide transmutation for over 40 years, with capabilities to generate source terms for accident analyses, characterize used fuel (including activity, decay heat, radiation emission rates, and radiotoxicity), activate structural materials, and perform fuel cycle analysis studies. This wide range of applications is possible because the guiding principle has been to explicitly simulate all decay and neutron reaction pathways using the best available data and to rigorously validate the result versus experiment. As an integral part of SCALE 6.1, ORIGEN has been subject to hundreds of validation cases using measured data from destructive isotopic assay of spent fuel, decay heat of spent fuel, gamma spectra resulting from burst fission, and neutron spectra resulting from spontaneous fission and (α,n) reactions.

In 2013, the ORIGEN API required a runtime per solution of 1 s. For depletion time to be brief compared to transport runtime, a desired runtime of 0.1 s was proposed. To achieve this goal, the general-purpose burnup chain of ~2,200 nuclides and ~54,000 transitions (valid for any time scale and any material), was simplified to include only isotopes important for LWR core physics. The current simplified burnup chain “CASL2.0” contains the 263 nuclides and the total runtime is reduced by a factor of 10 while preserving quantities such as total energy production, activity, mass, and macroscopic cross sections.

2.4. Thermal expansion methodology

Although the geometric data is provided in the BEAVRS benchmark, it is all provided at room temperature conditions. The geometric changes that occur between room temperature and operating conditions in a pressurized water reactor (PWR) are significant and must be taken into account. This is achieved by thermally expanding the core plate, nozzles, grid spacers, and pin geometry to a user-specified temperature. A linear expansion model:

$$\Delta L(T) = L(T_0)\int_{T_0}^{T}\alpha(T')dT', \tag{1}$$

is used to describe the dimensional change for each component based on its material. The methodology used to thermally expand is explained in Palmtag et al. (2017). For this analysis, all temperatures are thermally expanded to 565 K. While this is not exact for every condition simulated in this analysis, getting to hot conditions covers the majority of the effect.

2.5. Fuel temperature tables

The behavior of fuel temperature is expected to change with burnup and power. The BISON (Hales et al., 2013) fuel performance code is part of the VERA environment, but it is not computationally feasible to be run when tightly integrated with core follow analysis. Instead, BISON is used to generate fuel temperatures across a wide range of powers and burnups (Stimpson, 2016). The fuel temperatures are then fitted using a least-squares regression to obtain a quadratic relationship between the fuel temperature and the power at each burnup point:

$$T_{fuel}(P, Bu) = T_{mod} + a(Bu)P + b(Bu)P^2. \quad (2)$$

MPACT linearly interpolates the coefficients of the quadratic fit to the exact burnup point and then applies the quadratic relationship to obtain the fuel temperature during the solve.

3. Model development

The core geometry for cycles 1 and 2 is described in great detail in the benchmark specification. The model attempts to stay as faithful to the benchmark geometry as possible. The model is built to the benchmark specification, and it captures the detail of the fuel, gap, and clad in every fuel pin; explicit modeling of the PYREX burnable inserts (including detailed axial and radial description); explicit modeling of the control rods (including end caps, plenums, and hybrid rod geometry); and explicit modeling of the radial reflector (including baffle, barrel, and neutron pads). Fig. 1 shows the 2D slice of the MPACT geometry used for cycle 1.

The radial reflector is modeled by adding the barrel and neutron pads and then truncating the model so that the reflector region accounts for a single assembly pitch. The effect of this approximation was recently studied (Stimpson et al., 2017) and the effect of using the truncated model on core eigenvalue as <1 pcm and the maximum difference in pin power was $< 0.1\%$.

4. Cycle 1 results

The cycle 1 zero power physics tests include five critical measurements with several different control rod configurations. Table 1 shows the calculated boron concentration for all of the critical configurations.

All five of the measured critical configurations show good agreement with the calculated critical configurations, and they also agree well with previously published solutions (Kelly et al., 2014; Ryu et al., 2014). When more control rod banks are inserted into the core, the solutions show larger differences than when only a few banks are inserted. Although the agreement with previously published solutions is comparable, significant modifications to the control rod description in the updated benchmark specification (v. 2.0.1) make direct comparisons to other calculated results difficult.

In addition to the five critical conditions, the control rod reactivity worth is measured for all banks. This was done by modeling the control rod in and out with constant boron. For control rods for

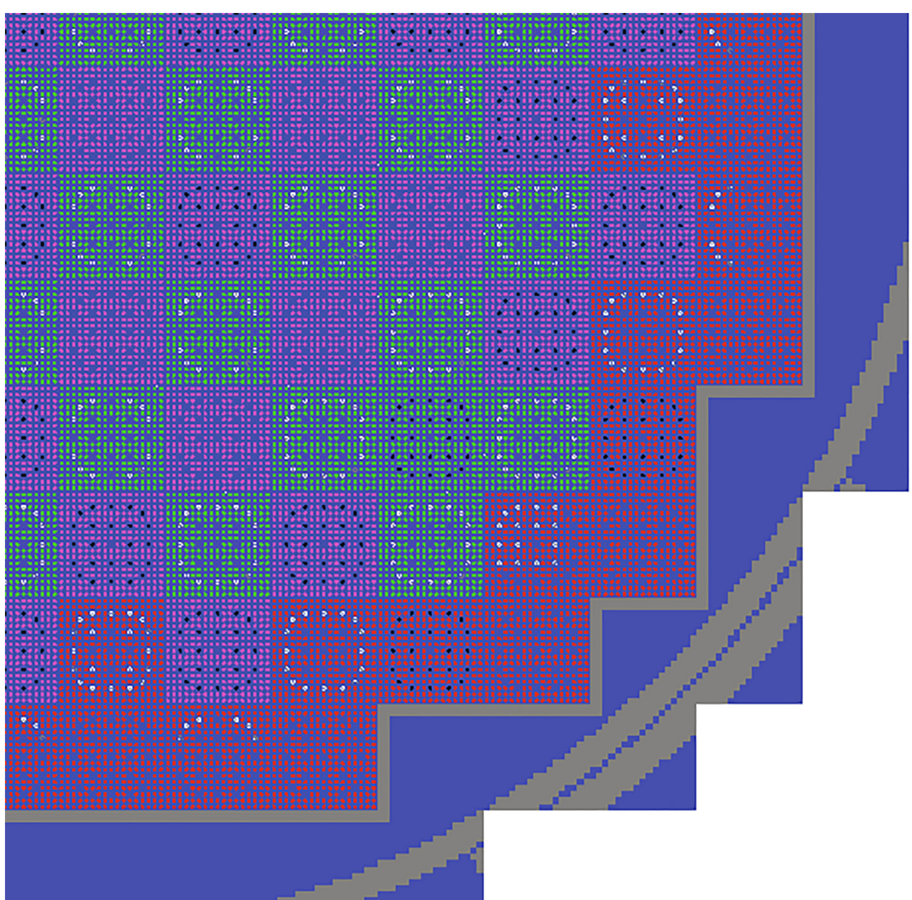

Fig. 1. BEAVRS core geometry.

Table 1
Cycle 1 zero power critical boron comparisons.

	Measured [ppm]	Calculated [ppm]	Difference [ppm]
ARO	975.0	975.9	0.9
D In	902.0	914.8	12.8
C/D In	810.0	817.1	7.1
A/B/C/D In	686.0	680.7	−5.3
A/B/C/D/SE/SD/SC In	508.0	491.9	−16.1

which critical boron concentrations were not given, the boron is interpolated based on the measured control rod worth. Table 2 shows the measured and predicted control rod worths. The predicted rod worth is less than 6% different for all banks.

The other measurement that can be compared is the calculation of the isothermal temperature coefficient (ITC). The calculation of ITC was performed by executing a 4-degree Fahrenheit perturbation to all of the temperatures above and below nominal conditions. The ITC is calculated by using the eigenvalue of both of these points and is displayed in Table 3. The ITC calculations all compare very well to the measured values, even for different control rod configurations.

Table 2
Cycle 1 zero power control rod worth.

	Measured [pcm]	Calculated [pcm]	Difference [%]
D In	788.0	779.3	−1.11%
C with D In	1203.0	1252.9	4.15%
B with C/D In	1171.0	1195.7	2.11%
A with B/C/D In	548.0	578.8	5.61%
SE with A/B/C/D In	461.0	487.7	5.80%
SD with SE/A/B/C/D In	772.0	779.4	0.96%
SC with SD/SE/A/B/C/D In	1099.0	1097.9	−0.10%
Total	6042.0	6171.8	2.15%

Table 3
Cycle 2 zero power isothermal temperature coefficient comparisons.

	Measured [pcm/F]	Calculated [pcm/F]	Difference [pcm/F]
ARO	−1.75	−2.56	−0.85
D In	−2.75	−4.01	−1.26
C/D In	−8.01	−8.80	−0.79

In addition to the zero power measurements, VERA is used to simulate the plant operation in cycle 1. During the first cycle, the cycle capacity factor was only 57%, and there were three long outages and several shorter outages. Even though day-by-day power histories are provided, the runtime for VERA would be relatively prohibitive to simulate the reactor at this level of detail. Instead, times and power levels were chosen that align with the times when flux maps were obtained during plant operation. Additionally, points were added to capture the 3 large outages during the cycle. Fig. 2 shows the power history for cycle 1 and the model power which was used in this simulation. Additionally, the black dots represent all flux map measurements that are available, and the green triangles represent each point where measured boron concentration is available.

The calculated boron concentration throughout the cycle is shown in Fig. 3.

Two different boron concentrations are reported in the specification. The first is the boron concentration corrected to 100% power with all rods out at regular intervals throughout the cycle. The second set is the instantaneous boron concentration at each flux map. Since the VERA model depletes with all rods out, the results are more consistent with the corrected boron concentration, although the boron concentrations at lower power levels should be expected to vary. The boron concentration agrees at the beginning of the cycle, but the difference grows as the cycle progresses. Fig. 3 shows the difference from measured boron when the reactor is close to full power.

In addition to the critical boron concentration, 24 flux maps are provided at various points throughout cycle 1 and 16 are chosen for comparison. The other 8 are excluded because they are all from the initial power ramp to full power, and sufficient data is not provided in the benchmark to accurately model Xenon during this time frame. A few other flux maps show larger discrepancies that are not at equilibrium conditions during startup or shortly after a power maneuver, but the data is included in the analysis.

The detector responses are extracted using the detector model described above, and they are normalized so they are consistent with the measured data. Since the mesh in the model is different than the 61 level measured data, the predicted detector signal is fit with a cubic spline and integrated onto 61 evenly spaced regions. Each detector string is then compared against the detector data.

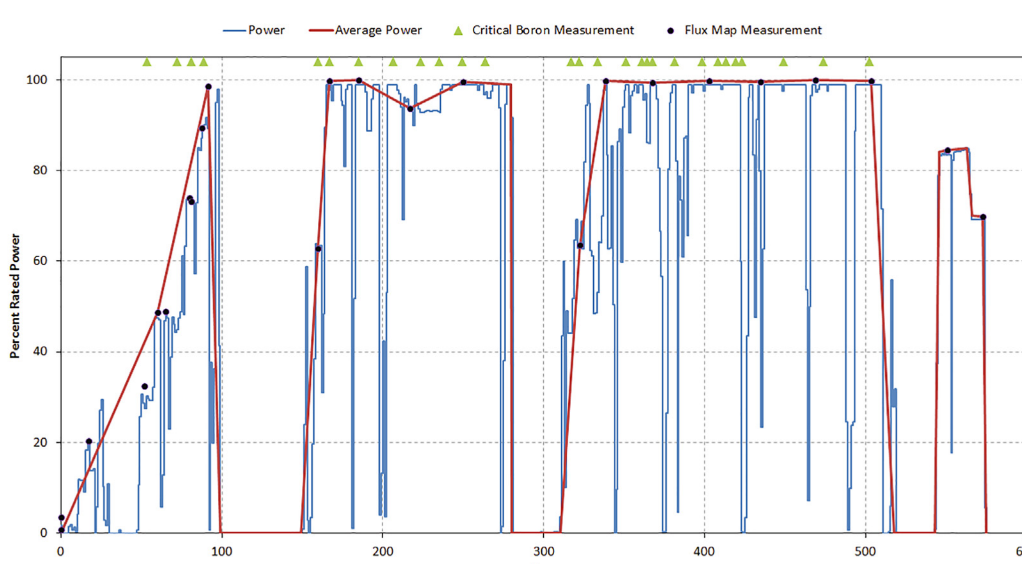

Fig. 2. Cycle 1 operating history.

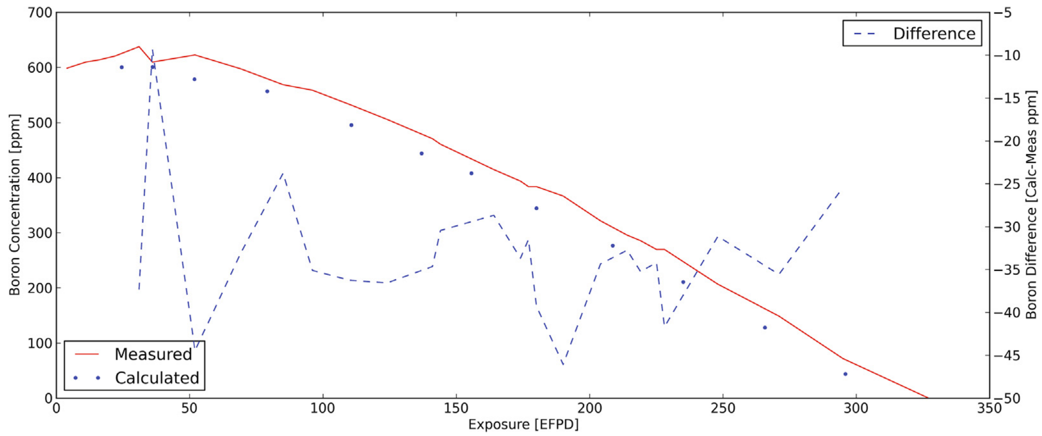

Fig. 3. Cycle 1 boron letdown comparison.

Three metrics are used to determine the quality of the simulation for the flux maps. The first is the 3D root mean square (RMS) comparison. This is simply the RMS of the difference in detector responses for all detector locations (*det*) and every axial level (*z*):

$$RMS_{3D} = \left(\frac{\sum_{det} \sum_{N_z} \left(\gamma^{meas}_{det,z} - \gamma^{calc}_{det,z} \right)^2}{N_{det} N_z} \right)^{\frac{1}{2}} \tag{3}$$

The second is the 2D, or radial, RMS which, is the difference of the axially integrated detector response:

$$RMS_{2D} = \left(\frac{\sum_{det} \left(\sum_{N_z} \gamma^{meas}_{det,z} - \sum_{N_z} \gamma^{calc}_{det,z} \right)^2}{N_{det} N_z} \right)^{\frac{1}{2}} \tag{4}$$

The final metric is the difference in axial offset as calculated by the incore detectors:

$$\Delta_{AO} = \frac{\sum_{det} \left(\frac{\sum_{z=N_z/2+1}^{N_z} \gamma^{meas}_{det,z} - \sum_{z=1}^{N_z/2} \gamma^{meas}_{det,z}}{\sum_{z=1}^{N_z} \gamma^{meas}_{det,z}} - \frac{\sum_{z=N_z/2+1}^{N_z} \gamma^{calc}_{det,z} - \sum_{z=1}^{N_z/2} \gamma^{calc}_{det,z}}{\sum_{z=1}^{N_z} \gamma^{calc}_{det,z}} \right)}{N_{det}} \tag{5}$$

Flux maps throughout the cycle are compared and analyzed using a python post processing script. The BEAVRS core has 58 locations where U-235 fission chambers measure the axial fission rate at 61 equidistant axial levels. During operation it is not uncommon to have inoperable detector locations which are excluded from data processing. To increase the number of detectors that can be compared, the VERA solution is unfolded from quarter core to the full core geometry using rotational symmetry and detector traces are compared at every available detector location. In order to display results, the comparisons are mapped back to the southeast quadrant using the same symmetry. In the case where multiple traces are present for symmetric locations, all detector signals are shown. Fig. 4 shows the output of this python script, along with the statistics discussed above.

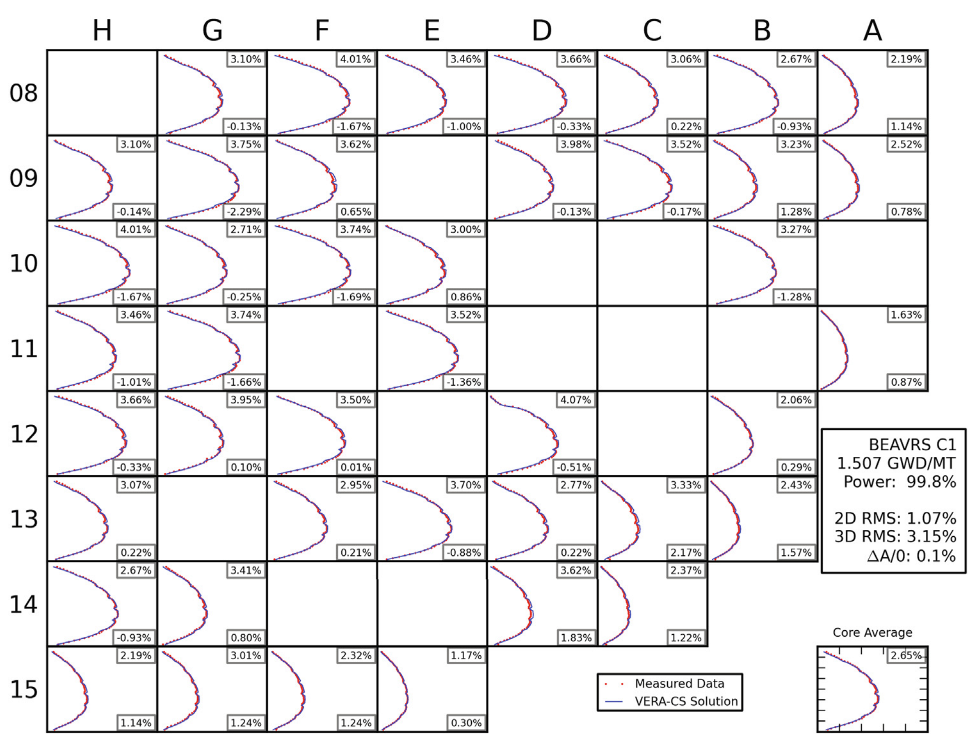

Fig. 4. Cycle 1 flux map at 1.507 GWD/MT.

Table 4
Cycle 1 flux map comparisons.

Exposure [GWD/ MT]	Power [%]	2D RMS [%]	3D RMS [%]	ΔAO [%]
1.023	98.70	1.67	4.55	0.46
1.296	62.80	3.42	5.16	−1.68
1.507	99.80	0.84	3.02	0.42
2.163	100.00	1.18	3.11	0.80
3.297	93.80	0.82	3.76	2.26
4.614	99.60	0.90	3.40	−0.50
6.012	63.60	1.18	4.53	−2.31
6.490	99.70	1.23	5.28	3.03
7.508	99.30	0.88	4.17	2.16
8.701	99.90	0.99	4.11	1.73
9.803	99.50	3.24	5.19	0.60
11.084	99.90	1.12	4.26	−0.32
12.343	99.80	1.21	4.10	−0.91
12.915	84.50	1.48	4.90	2.12
Cycle Average		1.44	4.25	0.56

The script is used to extract the 2D RMS, 3D RMS, and difference in axial offset for all available flux maps. Table 4 summarizes the comparisons for all of the flux maps. Of the remaining flux maps, the radial RMS is between 1 and 2%, and the 3D RMS is between 3 and 5%. In most cases, the comparison of axial offset is also less than 2.0%. There are several flux maps shown that exhibit higher-than-normal statistics. Specifically, two flux maps at around 60% power shows data during a power ascension and are potentially not at equilibrium Xenon conditions. Another data point which is slightly elevated is at 9.803 GWD/MT. This data point occurs very soon after a small outage that was not modeled and equilibrium conditions may not have been reached.

At the end of cycle, the flux maps also demonstrate excellent agreement which can be seen in Fig. 5. Fig. 5 shows that VERA predicts the axial and radial power throughout the reactor well. While there are some large discrepancies, the cycle 1 benchmark results are acceptable considering the assumptions made in the power history for this cycle.

5. Cycle 2 results

The detailed isotopic distributions obtained in the simulation of cycle 1 are shuffled throughout the core to obtain the initial model for cycle 2. Fig. 6 shows the pin exposure for the core after the shuffle. The dark blue locations represent fresh fuel loaded into the core for cycle 2. Significant gradients in exposure on the fuel can be observed; current design tools cannot accurately predict these gradients. In addition to being shuffled the fuel was decayed for three months. This value is not prescribed by the benchmark, so three months was assumed.

The zero power physics tests for cycle 2 are simulated with the two provided critical boron concentrations and are compared in Table 5.

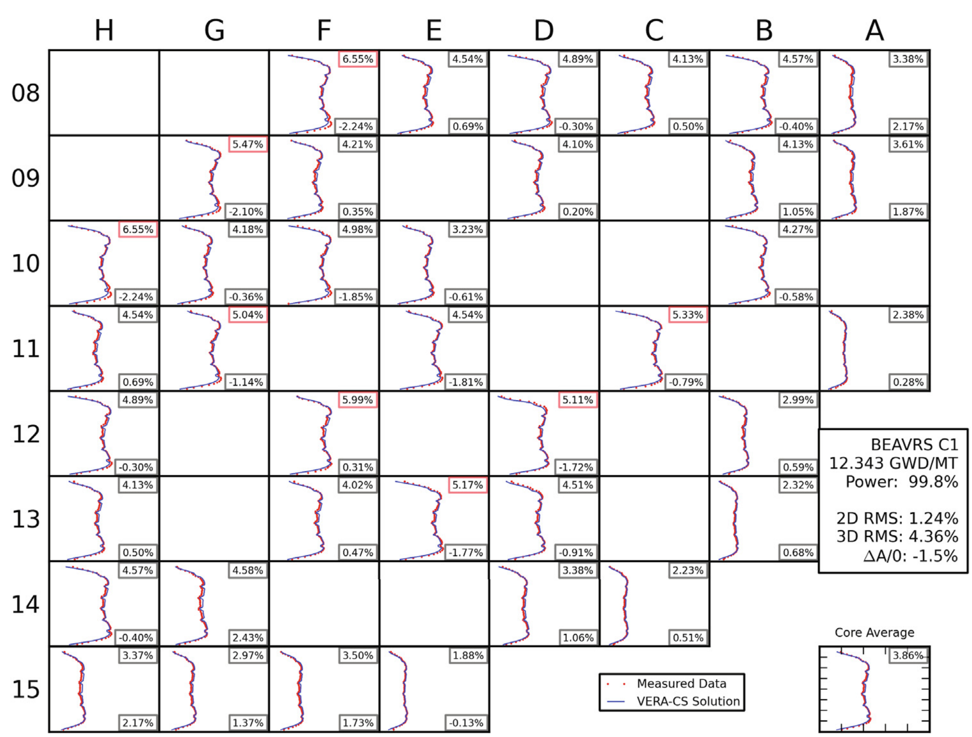

Fig. 5. Cycle 1 flux map at 12.343 GWD/MT.

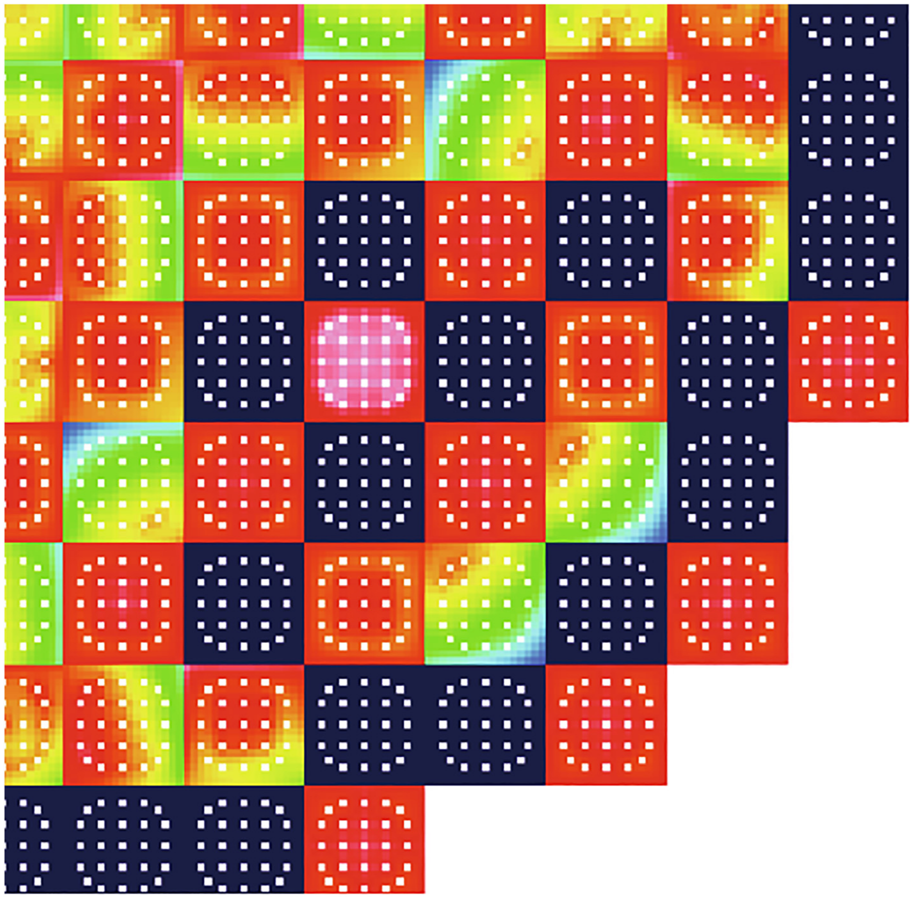

Fig. 6. Cycle 2 beginning of cycle (BOC) pin exposures [GWD/MT].

Table 5
Cycle 2 zero power critical boron comparisons.

	Measured [ppm]	Calculated [ppm]	Difference [ppm]
ARO	1405.0	1396.4	−8.6
C In	1273.0	1294.4	21.4

The cycle 2 power history is much smoother than the power history for cycle 1. The model is simulated using a burnup-average power for each time step. Fig. 7 shows the day-by-day power history and the model power used. As in cycle 1, the flux map points were chosen to allow for direct comparisons with the measured data.

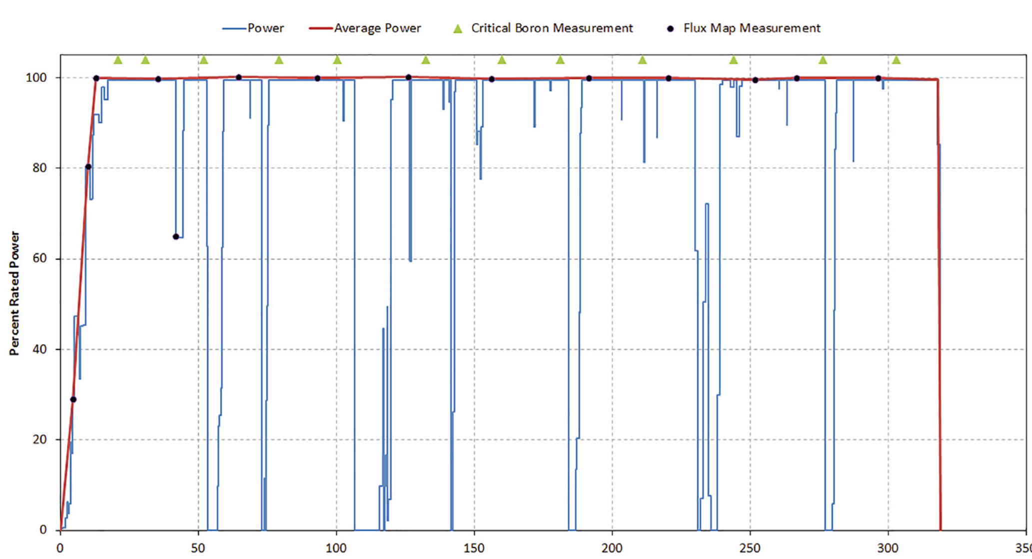

Fig. 7. Cycle 2 operating history.

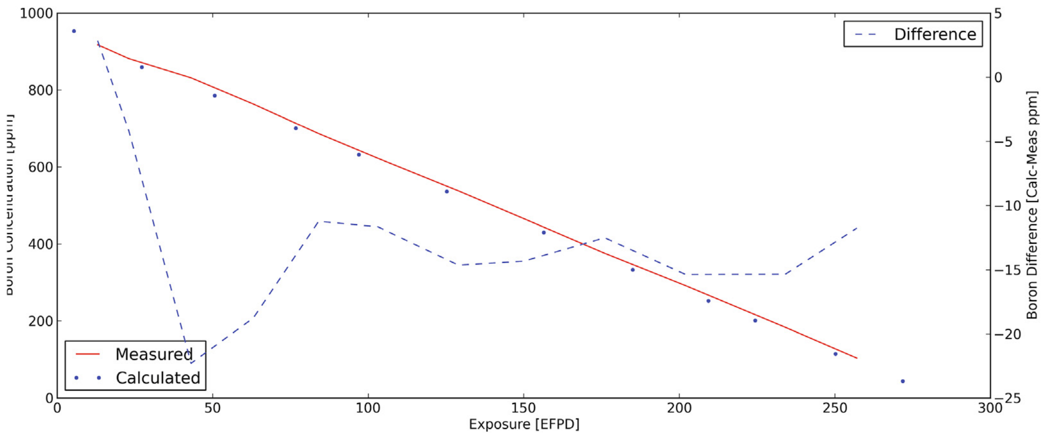

Fig. 8. Cycle 2 boron letdown comparison.

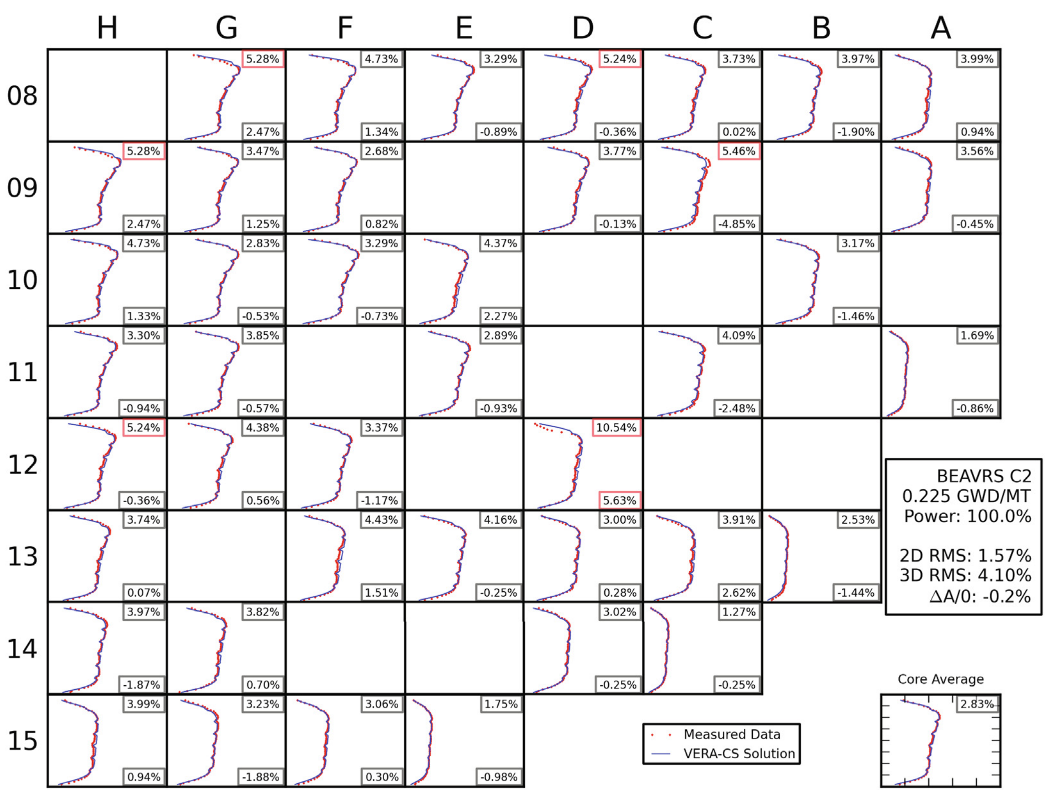

Fig. 9. Cycle 2 flux map at 0.225 GWD/MT.

Fig. 8 shows the boron letdown curve for the simulation of cycle 2. The boron predictions are consistently 16 ppm low, which is a considerably better than cycle 1.

The flux map comparisons are also made for cycle 2. The first flux map at 100% is shown in Fig. 9. For this case, the power is significantly peaked toward the top of the core, but VERA predicts the axial power shape very well. Standard operating practice for similar reactors to leave Bank D inserted 10–20 steps during operation to provide sufficient control rod worth to account for depletion effects on reactivity between critical boron updates. Since the BEAVRS specification does not provide the location of Bank D, it is not inserted in this model. Location D12 shows signs of an over-prediction in detector signal likely due to Bank D not being present.

The detector comparisons for cycle 2 also show good comparisons throughout the cycle, as seen in Table 6. With the exception of the first two flux maps which are taken during the initial

Table 6
Cycle 2 flux map comparisons.

Exposure [GWD/MT]	Power [%]	2D RMS [%]	3D RMS [%]	ΔAO [%]
0.013	29.10	3.11	5.57	2.09
0.126	80.50	3.22	7.58	−3.91
0.225	100.00	1.57	4.10	−0.24
1.128	99.70	1.11	3.52	0.29
2.095	100.00	1.04	3.54	−0.48
3.175	99.90	1.15	4.39	−2.13
4.013	100.00	1.14	3.76	−1.56
5.187	99.80	1.12	4.18	−1.83
6.475	99.90	1.18	3.97	−1.90
7.658	100.00	1.38	3.83	−1.49
8.665	99.50	0.95	4.49	−2.13
9.287	99.90	1.16	4.15	−1.68
10.356	99.90	1.11	3.49	−0.82
Cycle Average		1.43	4.15	−0.68

escalation of power, the flux map comparisons are as good as they were in cycle 1, and there are no discernible trends in the behavior throughout the operation of cycle 2.

The simulation of cycle 1 and 2 were perform on 880 Intel Xeon 2.3 GHz processors with 4 GB of RAM per processor. The depletion of cycle 1 included 28 state points and took 24 h and 50 min of total wall-time. Cycle 2 only has 16 states due to its simpler power history and used 13 h and 26 min of wall time. Overall, VERA averaged about 50 min (760 CPU-hrs) per state point for these calculations.

6. Conclusions

The BEAVRS benchmark has been successfully completed with the VERA code suite. State-of-the-art computer codes have been tightly coupled and integrated into a core simulation capability providing many of the features needed to simulate PWRs. Models were developed that stayed faithful to the benchmark geometry. A power history is approximated for cycles 1 and 2 which captures the major components of the power history defined by the benchmark. Zero power physics tests showed very small biases in the results. Cycle 1 core follow simulations showed a 30 ppm underprediction of boron concentration throughout the cycle. It is expected that the complicated power history in cycle 1 led to the difficulty in predicting the results. However, even though there were issues with core reactivity, the flux map comparisons throughout cycle 1 were very good. The cycle 2 core follow results demonstrated a 16 ppm under prediction of boron concentration throughout the cycle, which is much improved from cycle 1. The flux map comparisons for cycle 2 also demonstrate very good comparisons with measured data.

CRediT authorship contribution statement

Benjamin Collins: Methodology, Software, Visualization, Formal analysis, Writing - original draft. **Andrew Godfrey:** Formal analysis, Resources. **Shane Stimpson:** Software. **Scott Palmtag:** Software, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, athttps://doi.org/10.1016/j.anucene.2020.107602.

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